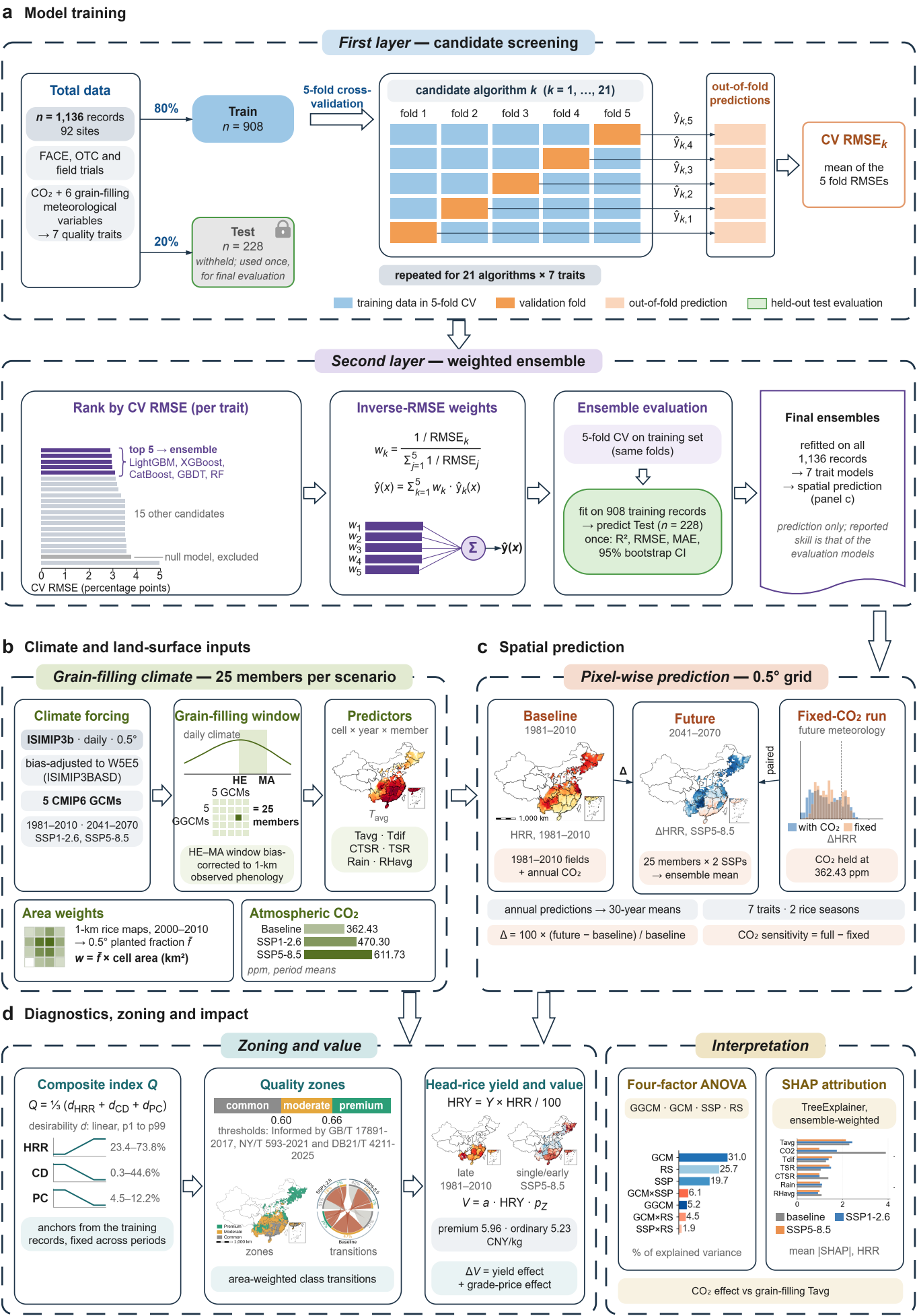

a Model training
First layer — candidate screening
Total data
n = 1,136 records
92 sites
FACE, OTC and field trials
CO₂ + 6 grain-filling meteorological variables
→ 7 quality traits
80%
Train
n = 908
20%
Test
n = 228
withheld; used once, for final evaluation
5-fold cross-validation
candidate algorithm k (k = 1, …, 21)
fold 1
fold 2
fold 3
fold 4
fold 5
out-of-fold predictions
CV RMSE_k
mean of the 5 fold RMSEs
repeated for 21 algorithms × 7 traits
training data in 5-fold CV
validation fold
out-of-fold prediction
held-out test evaluation
Second layer — weighted ensemble
Rank by CV RMSE (per trait)
top 5 → ensemble
LightGBM, XGBoost, CatBoost, GBDT, RF
15 other candidates
null model, excluded
CV RMSE (percentage points)
Inverse-RMSE weights
Ensemble evaluation
5-fold CV on training set (same folds)
fit on 908 training records → predict Test (n = 228) once: R², RMSE, MAE, 95% bootstrap CI
Final ensembles
refitted on all 1,136 records
→ 7 trait models
→ spatial prediction (panel c)
prediction only; reported skill is that of the evaluation models
b Climate and land-surface inputs
Grain-filling climate — 25 members per scenario
Climate forcing
ISIMIP3b · daily · 0.5°
bias-adjusted to W5E5 (ISIMIP3BASD)
5 CMIP6 GCMs
1981–2010 · 2041–2070
SSP1-2.6, SSP5-8.5
Grain-filling window
daily climate
HE
MA
5 GCMs
5 GGCMs
= 25 members
HE–MA window bias-corrected to 1-km observed phenology
Predictors
cell × year × member
Tavg · Tdif
CTSR · TSR
Rain · RHavg
Area weights
1-km rice maps, 2000–2010
→ 0.5° planted fraction f̄
w = f̄ × cell area (km²)
Atmospheric CO₂
Baseline 362.43
SSP1-2.6 470.30
SSP5-8.5 611.73
ppm, period means
c Spatial prediction
Pixel-wise prediction — 0.5° grid
Baseline
1981–2010
1,000 km
HRR, 1981–2010
1981–2010 fields + annual CO₂
Δ
Future
2041–2070
ΔHRR, SSP5-8.5
25 members × 2 SSPs → ensemble mean
paired
Fixed-CO₂ run
future meteorology
with CO₂
fixed
ΔHRR
CO₂ held at 362.43 ppm
annual predictions → 30-year means
7 traits · 2 rice seasons
Δ = 100 × (future − baseline) / baseline
CO₂ sensitivity = full − fixed
d Diagnostics, zoning and impact
Zoning and value
Composite index Q
Q = ⅓ (d_HRR + d_CD + d_PC)
desirability d: linear, p1 to p99
HRR 23.4–73.8%
CD 0.3–44.6%
PC 4.5–12.2%
anchors from the training records, fixed across periods
Quality zones
common
moderate
premium
0.60
0.66
thresholds: Informed by GB/T 17891-2017, NY/T 593-2021 and DB21/T 4211-2025
zones
transitions
area-weighted class transitions
Head-rice yield and value
HRY = Y × HRR / 100
late 1981–2010
single/early SSP5-8.5
V = a · HRY · p_Z
premium 5.96 · ordinary 5.23 CNY/kg
ΔV = yield effect + grade-price effect
Interpretation
Four-factor ANOVA
GGCM · GCM · SSP · RS
GCM 31.0
RS 25.7
SSP 19.7
GCM×SSP 6.1
GGCM 5.2
GCM×RS 4.5
SSP×RS 1.9
% of explained variance
SHAP attribution
TreeExplainer, ensemble-weighted
Tavg
CO2
Tdif
TSR
CTSR
Rain
RHavg
baseline
SSP1-2.6
SSP5-8.5
mean |SHAP|, HRR
CO₂ effect vs grain-filling Tavg